Choose Inputs and Outputs

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

One connected input/output and baseline record.

State activity requirements, trace requests, distinguish missing from zero and complete the independent room baseline.

Keep one habit: Keep input validity, fitting logic, output and comparison conditions visible.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Find a Real Problem. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Verify Important **Output**s. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Find a Real Problem

Verify Important Outputs

Privacy by Choosing Less

W1 · State the complete activity requirements

> Use the supplied catalogue and explain the quiet condition precisely.

Your evidence record

- Activity reference for W1–W3: valid available time is whole minutes 0–30 and Quiet required is Yes or No. In order: B1 Bookmark drawing (10 min, quiet Yes); B2 Story circle (8 min, quiet No); B3 Pattern puzzle (12 min, quiet Yes); B4 Tower challenge (15 min, quiet No). Choose the first activity taking no more than the available time and meeting the quiet requirement. Quiet required No allows either activity type. Clarify invalid/missing inputs; report no match only for a valid request with no fitting activity. All materials are assumed available in this paper model.

- Replace “suggest a good activity” with the time condition, quiet condition and **tie rule** used in this project. Explain why Quiet required No permits B1 as well as non-quiet activities when their times fit.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Run three baseline requests

> Show the output and deciding checks for each.

Your evidence record

- Using the catalogue, record the **baseline output** and reason for 10 minutes with quiet required, 8 minutes with quiet not required, and 7 minutes with quiet not required. Identify which no-match result is justified and why.

My reasoning and evidence

MAIN LESSON · SECTION 14

W3 · Distinguish blank from valid zero

> Explain clarification versus a genuine no-match result.

Your evidence record

- Compare a request with blank minutes and Quiet required Yes to a request with 0 minutes and Quiet required Yes. What should the **baseline** record for each?

  Explain why these are different outcomes even though neither produces an activity suggestion.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Plan a fair baseline/prototype comparison

> Use the same requests, catalogue and judging rules, and retain all results.

Your evidence record

- Team A tests a prototype on five easy requests and the manual **baseline** on five different requests, then reports only the prototype's successful **output**s. Identify two changes needed for a fair comparison.

  If both methods then meet all **requirement**s, what can and cannot be concluded about compatibility?

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Build a baseline for a different catalogue

> Keep one first attempt before hints or feedback.

Read and explain independently

A paper room recommender uses R1 first, capacity 6, quiet Yes; then R2, capacity 10, quiet No. Valid requests give a whole group size from 1 to 12 and Quiet required Yes or No.

A room fits when capacity is at least group size and the quiet condition is met; choose the first **fitting** room. Record **output**s and reasons for 6 people Yes, 8 people No and 8 people Yes.

Explain how a manual **baseline** can be compared fairly with a prototype. No real booking is allowed.

> **My independent explanation, trace and limits**

Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Build a baseline for a different catalogue

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

Your evidence record

Freeze the sorter's v1 **input** list, **output contract**, and not-sure behavior.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Run the **baseline** on three requests for **Requirement**s and a simple **baseline**.
Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for **Requirement**s and a simple **baseline**. Proposed explanation: “Quiet required No excludes quiet activities”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Design a paper/plastic sorter that receives an object description. Include a mixed paper-and-plastic cup. Specify **input**s, **output**s and the uncertain-case action. Is the owner's name useful?

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

A sorter suggests paper or plastic trays using material **observation**s.

Name an **input**, **output** and unnecessary **input**.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

A paper sorter must suggest pencil tray or book shelf. Choose one needed **input** and state the **output**. What should happen if the picture is blocked?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 30 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Design a paper/plastic sorter that receives an object description. Include a mixed paper-and-plastic cup. Specify **input**s, **output**s and the uncertain-case action. Is the owner's name useful?

> **My explanation after practice**

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

Use this when helpful

A paper room recommender uses R1 first, capacity 6, quiet Yes; then R2, capacity 10, quiet No. Valid requests give a whole group size from 1 to 12 and Quiet required Yes or No.

A room fits when capacity is at least group size and the quiet condition is met; choose the first **fitting** room. Record **output**s and reasons for 6 people Yes, 8 people No and 8 people Yes.

Explain how a manual **baseline** can be compared fairly with a prototype. No real booking is allowed.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

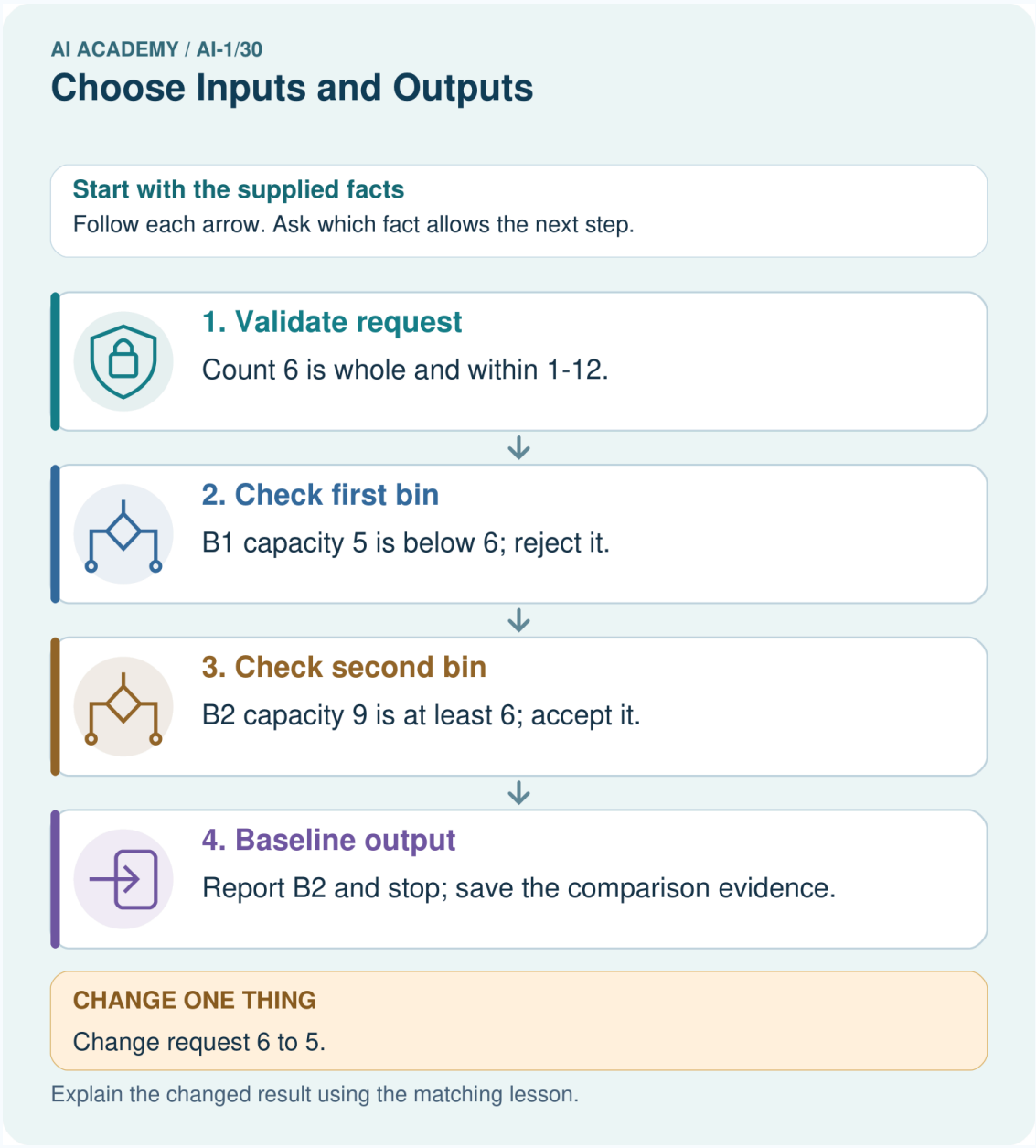
> Keep the explanation and the support you used.

Use this when helpful

A fictional classroom chooses a storage bin for model parts.

Catalogue order: B1 capacity 5, B2 capacity 9. Valid request is a whole part count 1–12.

Return first bin with capacity at least the count; none if neither fits; clarify invalid **input**.



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace requests 10 and blank. Why should they not receive the same explanation?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Document sorter schema, catalogue order, acceptance rule and all **output** types; use a manual trace as the comparison **baseline**.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Helpful Classroom Sorter

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Document sorter schema, catalogue order, acceptance rule and all **output** types; use a manual trace as the comparison **baseline**.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional classroom chooses a storage bin for model parts.
- Which labelled step matches this detail? Report B2 and stop; save the comparison **evidence**.
- 1. **Baseline output**
- 2. Check first bin
- 3. Validate request
- 4. Check second bin
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: **Baseline output** Report B2 and stop; save the comparison **evidence**. The **baseline** is an ordered catalogue scan, not a preference ranking.

It makes expectations testable before a prototype is built. Count 10 is valid but has no matching bin; that differs from a blank count, which fails the **input** contract. Both situations need explicit **output** behavior.

Changed-case question

- Changed-case practice: Change request 6 to 5. Which conclusion is supported by the supplied facts?
- 1. The **baseline** returns B1.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The **baseline** returns B1. Its capacity equals the request, and at least includes equality.

Repair and save your work

Review Find a Real Problem (ai-1/29). Goal: Describe a specific user difficulty using supplied **evidence** without inventing its cause.

Short practice: Write a one-sentence project goal and say what **evidence** would count as a failure.. Return to this week and explain what you changed.

Review Verify Important **Output**s (ai-1/27). Goal: Separate an answer into claims and connect each claim to relevant **evidence**.

Short practice: Correct the draft and identify which part needs another source rather than a calculation.. Return to this week and explain what you changed.

Review Privacy by Choosing Less (ai-1/24). Goal: Justify which information a stated task needs and remove unnecessary fields.

Short practice: For the original task, does adding an anonymous-looking pupil code improve the repair count?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

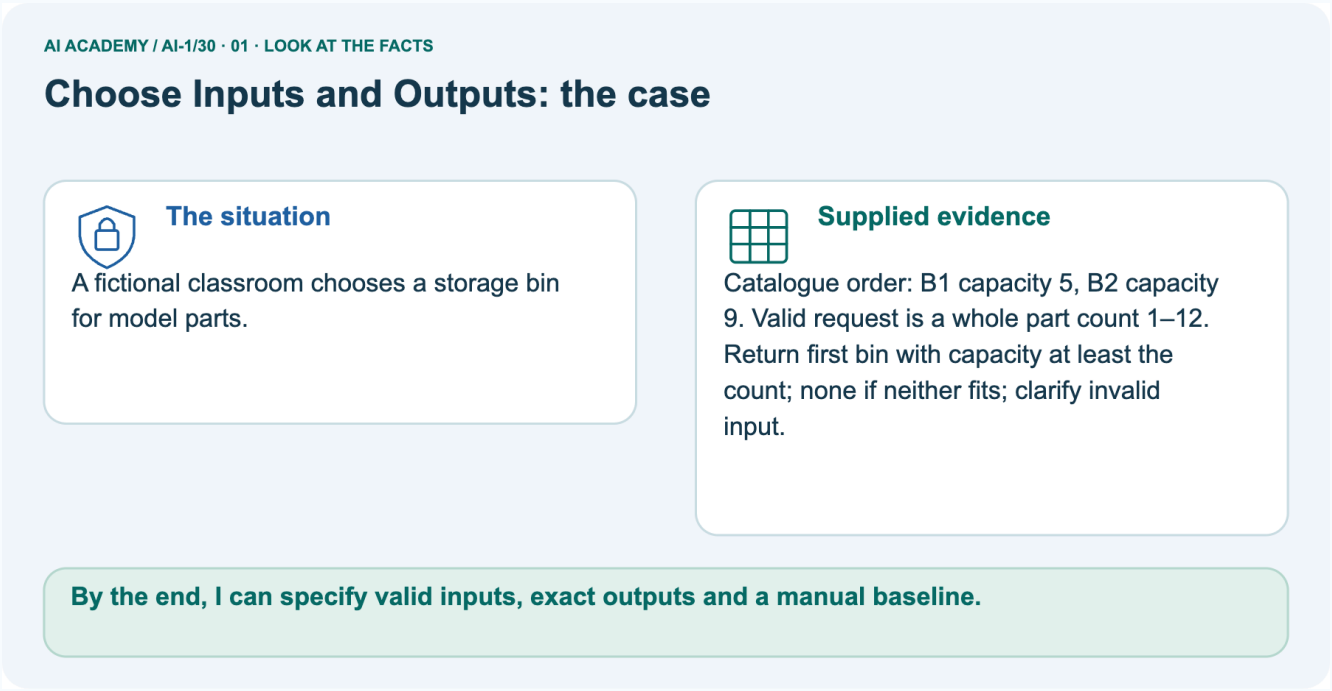
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Choose Inputs and Outputs: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

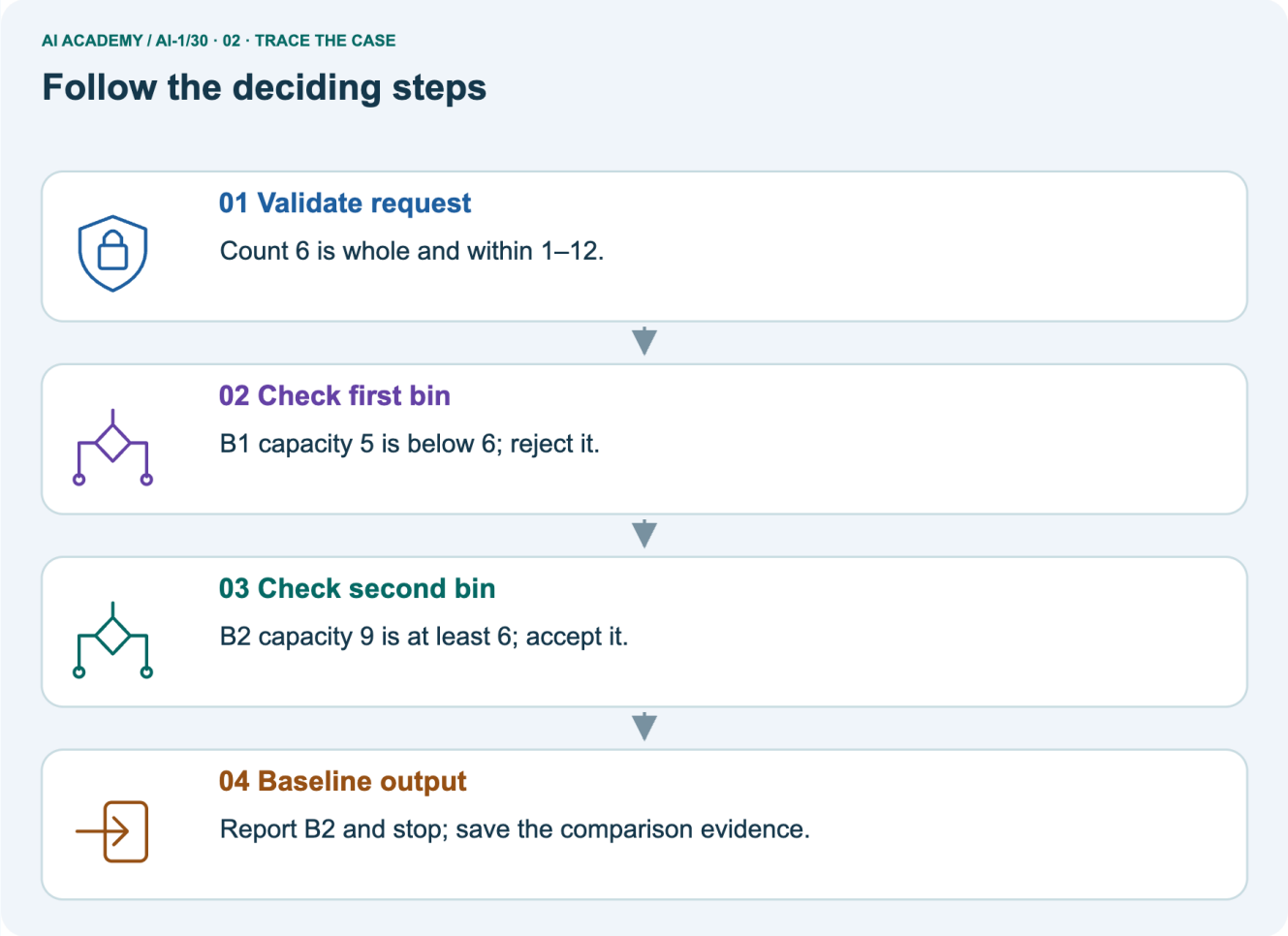
A fictional classroom chooses a storage bin for model parts. Supplied facts: Catalogue order: B1 capacity 5, B2 capacity 9. Valid request is a whole part count 1–12.

Return first bin with capacity at least the count; none if neither fits; clarify invalid **input**.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Validate request: Count 6 is whole and within 1–12. → Check first bin: B1 capacity 5 is below 6; reject it. → Check second bin: B2 capacity 9 is at least 6; accept it. → **Baseline output**: Report B2 and stop; save the comparison **evidence**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. The manual checklist baseline

AI ACADEMY / AI-1/30 · 03 · INSPECT THE LESSON MODEL

The manual checklist baseline

Stage	Manual action	Recorded result
Validate the request	Check minutes and Yes or No	Clarify invalid input
Inspect catalogue in order	Check time and quiet for each	Mark compatible rows
Choose or explain absence	First compatible row or none	Suggestion or no match
Keep the evidence	Record input, result, reason	Comparable baseline record

The baseline is a complete manual method. A prototype must be compared with this defined procedure, not an invented weak opponent.

Follow the labelled picture with the matching explanation. Open the vector diagram

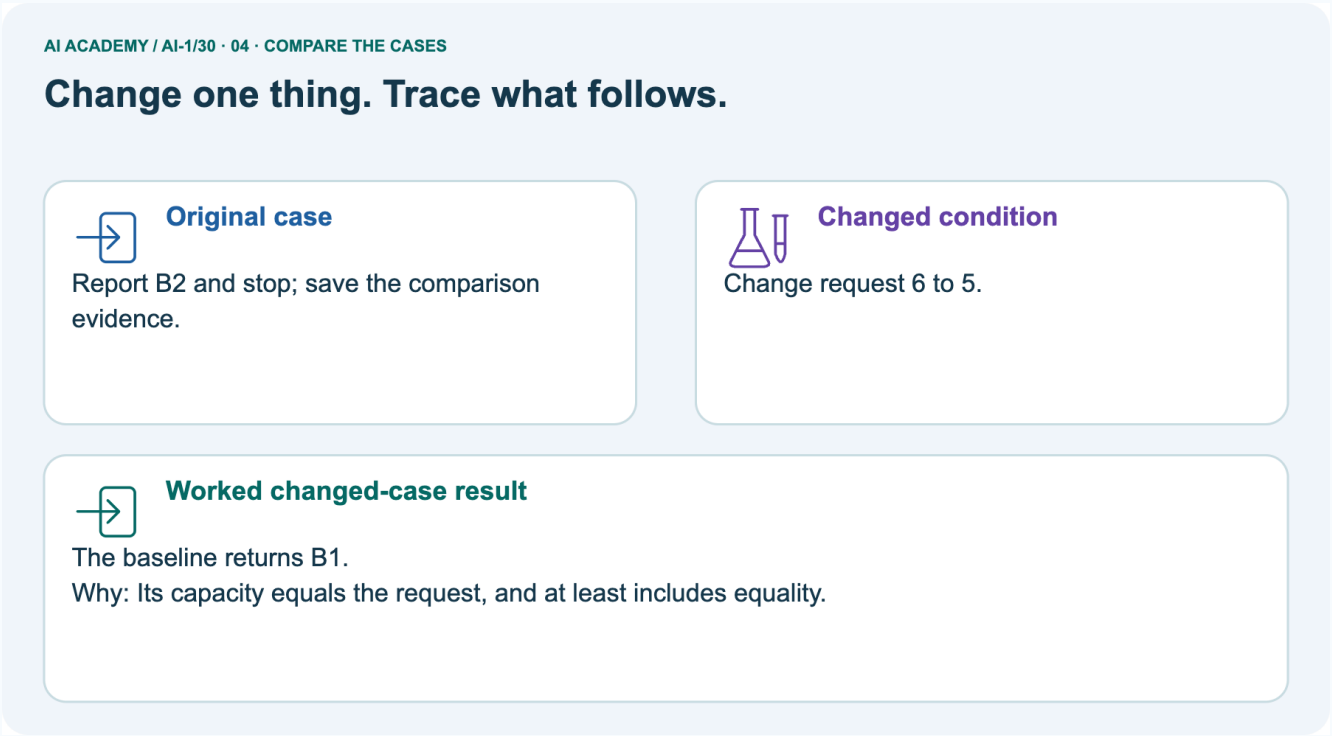
The manual checklist **baseline**. Validate the request | Check minutes and Yes or No | Clarify invalid **input**. Inspect catalogue in order | Check time and quiet for each | Mark **compatible** rows.

Choose or explain absence | First **compatible** row or none | Suggestion or no match. Keep the **evidence** | Record **input**, result, reason | Comparable **baseline** record

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



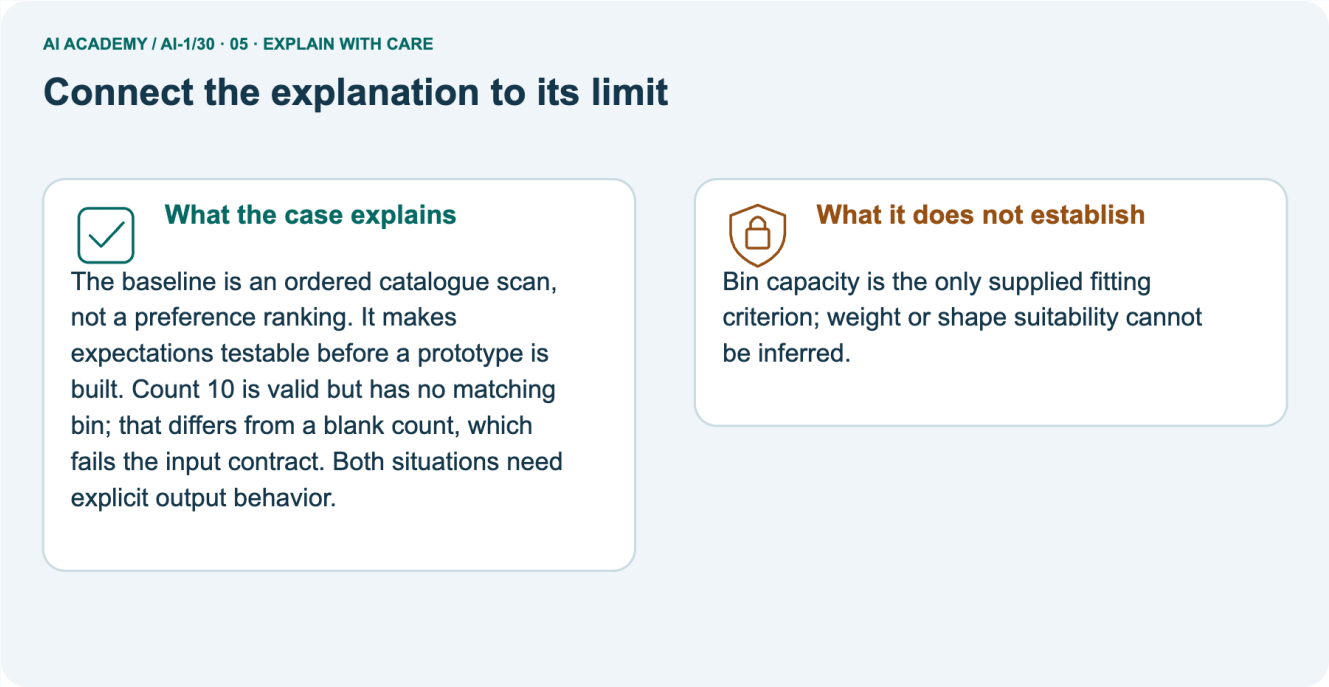
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Report B2 and stop; save the comparison **evidence**. Changed condition: Change request 6 to 5. Expected result: The **baseline** returns B1. Reason: Its capacity equals the request, and at least includes equality.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The **baseline** is an ordered catalogue scan, not a preference ranking. It makes expectations testable before a prototype is built.

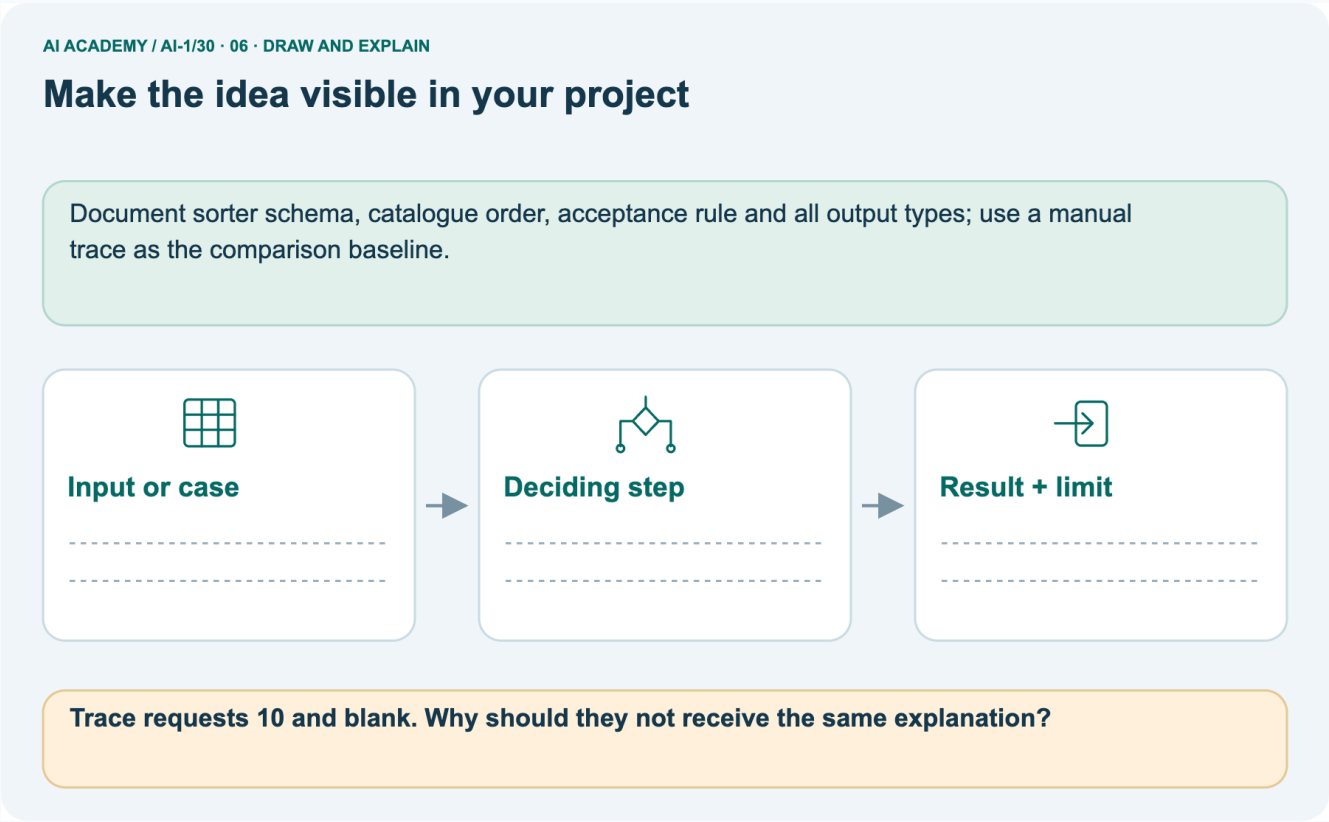
Count 10 is valid but has no matching bin; that differs from a blank count, which fails the **input** contract. Both situations need explicit **output** behavior.

Limit: Bin capacity is the only supplied **fitting** criterion; weight or shape suitability cannot be inferred.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document sorter schema, catalogue order, acceptance rule and all **output** types; use a manual trace as the comparison **baseline**. Draw your **input**, deciding step, result and limit. Supported question: Trace requests 10 and blank.

Why should they not receive the same explanation?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Choose Inputs and Outputs: the case

02. Follow the deciding steps

03. The manual checklist baseline

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.